The Human Cost of Machine Speed: AI, Trust, and Escalation in Modern Warfare

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The integration of artificial intelligence into military systems has transformed how war is waged and understood. AI-enabled drones, autonomous targeting algorithms, and decision-support systems now process data and recommend or even execute lethal actions faster than any human operator ever could. While these advancements promise efficiency, they also introduce new risks, especially the risk of accidental escalation. Scholars and defense analysts have begun warning of a new phenomenon called "flash wars," which are conflicts that unfold or intensify so rapidly that human intervention becomes impossible before catastrophe strikes (UNRIC 2025). Where judgment, hesitation, and empathy once played a part in the decision to kill, algorithmic processes now dominate the cognitive and moral space of warfare. As a result, human factors like trust, moral hesitation, and emotional restraint are increasingly replaced by machine-driven logic. The Harvard Medical School's "Risks of Artificial Intelligence in Weapons Design" report warns that, "If few soldiers die in offensive warfare, it weakens the association between acts of war and human cost, and it becomes politically easier to start wars, which, in turn, may lead to more death and destruction overall" (Harvard 2024). The more distant decision-makers become from the consequences of violence, the easier it becomes to execute it.

In this context, the challenge is not merely technological but ethical and existential: How can we ensure that human beings remain meaningfully in control when war itself is becoming too fast for humans to comprehend? This paper argues that overtrust in AI systems, along with the condensed time it takes to make military decisions, undermines human oversight and heightens the risk of accidental escalation. Preventing flash wars will require not only new governance mechanisms and institutional reforms but also emphasizing the importance of human moral judgment. This paper looks at the tension between speed and reflection, efficiency and ethics, machine logic and human conscience. By combining recent military analyses, ethical frameworks, and real-world case studies from Petrov's Cold War restraint to the Israeli "Lavender" targeting system, this study seeks to show that the greatest danger of AI warfare is not rebellion by machines but obedience by humans.

Artificial intelligence is increasingly attractive to modern militaries because it is able to do tasks that overwhelm human operators in modern warfare. There are massive volumes of data from satellites, drones, sensors, communications intercepts, and surveillance systems, far more than human analysts can process in real time. Generative artificial intelligence offers a way to manage this information overload by rapidly synthesizing data, identifying patterns, and producing usable outputs for planning and operations. Generative AI models are trained as foundation models through unsupervised or semi-supervised learning, allowing them to build structure from large, unlabelled datasets and apply that knowledge across multiple tasks (NVIDIA n.d.). While these systems do not understand or reason like humans, they are best at understanding information at scale, making them useful for intelligence analysis, logistics planning, and operational coordination.

Militaries are drawn to AI not simply because it is new, but because it promises operational advantages. AI systems can process sensor feeds continuously, identify trends or anomalies faster than human analysts, and generate recommendations at a speed that compresses traditional decision cycles. This means faster intelligence fusion, more rapid targeting decisions, and shorter kill chains. The war in Ukraine points to this shift. AI-enabled drones using satellite imagery and image-recognition software can identify potential targets, communicate with centralized systems, and strike with minimal delay, effectively surpassing the slower, human-intensive processes that historically governed intelligence meeting and force deployment (U.S. Army 2024). In conflicts where speed determines survival, these advantages are difficult for militaries to ignore.

The pressure to adopt AI is intensified by historical experience. Military power has repeatedly shifted in favor of actors who embraced disruptive technologies early, while those who failed to adapt suffered strategic decline. The U.S. Army notes that innovations such as machine guns, tanks, and industrialized logistics fundamentally reshaped warfare and reordered global power structures (U.S. Army 2024). Russia's inability to industrialize its military rapidly

during the Industrial Revolution left it technologically outmatched by Germany during World War I, contributing to devastating losses despite its numerical strength. Once disruptive technologies redefine operational norms, late adopters face disadvantages. This logic has also shaped nuclear arms competition, where restraint was often viewed as vulnerability rather than a moral choice.

At the same time, AI is often poorly understood. Many of the most advanced machine-learning models used today are black box systems, meaning that even their creators cannot fully explain how internal decisions are produced. Deep-learning models rely on multilayered neural networks with hundreds or even thousands of hidden layers that transform inputs into outputs through complex mathematical operations (IBM 2023). While users, including AI developers, can observe what data enters the system and what predictions or classifications emerge, they cannot see what happens within the hidden layers in between. Developers may understand in general terms how information flows through a network, but they do not know precisely how the model combines internal representations to arrive at a given output. Even open-source models, which make their underlying code publicly available, remain opaque because the internal reasoning processes cannot be fully interpreted while the system is operating.

This opacity has serious consequences when errors or biases emerge. Hidden biases in training data can lead to unfair and harmful outcomes that are difficult to detect or correct. This issue was highlighted in the Apple Card gender bias controversy, in which users noticed that the algorithm appeared to offer significantly lower credit limits to women than to men, even when financial circumstances were similar. The case illustrates how AI systems trained on historical data can reproduce and amplify existing inequalities while presenting their decisions as neutral or objective. Since the model's internal logic was opaque, it was difficult to determine exactly which factors produced the biased outcomes or how to fix them. When a black box model produces inaccurate or harmful results, correcting the behavior is often challenging because

developers cannot pinpoint where the system is going wrong or which internal mechanisms need adjustment (IBM 2023).

While financial and military decisions may differ in usage, both rely on the same opaque models. In military contexts, this problem becomes even more dangerous. When black box systems are used for tasks such as threat identification or targeting, there may be no clear way to determine which variables led the system to classify an individual or location as hostile. The inability to trace the exact factors driving identification decisions undermines accountability and raises the risk that biased or flawed correlations will be translated directly into lethal action. Despite these risks, militaries continue to integrate AI because of the advantages it offers. Success increasingly depends on rapid information synthesis and compressed decision cycles, creating powerful incentives to deploy AI even when its internal logic remains opaque (U.S. Army 2024). Even though AI is not well understood, it offers clear military advantages. However, it may lead to a path of automation bias, overreliance on machine judgment, and the risk of rapid, accidental escalation.

Automation bias has become one of the most dangerous side effects of integrating artificial intelligence into warfare. Decades of psychological research show that humans have a natural tendency to trust automated systems, especially under conditions of uncertainty or stress. In the military context, this can be potentially fatal because operators begin to assume that the machine's recommendation is inherently more accurate than their own judgment. The Carnegie Council's "Ethics of AI Decision-Making in Conflicts" report explains that "humans tend to over-rely on AI recommendations and are prone to saliency bias, potentially skewing judgment rather than enhancing it" (Carnegie Council 2024). When soldiers or analysts are faced with time-sensitive data and must act within seconds, the appeal of machines can override critical reflection. This psychological dependency transforms AI from a tool into an authority. Instead of aiding human reasoning, it replaces it.

In battlefield environments where milliseconds separate deterrence from disaster, that misplaced trust can trigger escalation that no one intended. The Harvard Medical School's analysis reinforces this point by exposing the illusion of control embedded in so-called "human-in-the-loop" systems. The report warns that "another challenge is that tech companies are opaque about the degree of autonomy and human oversight in their weapons systems. For some, human oversight could mean pressing a 'go kill' button after an AI weapons unit makes a long chain of black box decisions, without the human understanding or being able to spot errors in the system's logic" (Harvard 2024). The act of pressing a single confirmation button is presented as moral supervision, but in reality, it often means the human is approving a machine's opaque reasoning without genuine comprehension. It becomes a dangerous illusion of ethical control where the interface disguises the depth of automation beneath it.

This oversight, performed under immense time pressure, strips operators of their ability to intervene meaningfully. The Carnegie report describes this as a cognitive trap in which "humans believe they are monitoring AI when in fact AI is monitoring them" (Carnegie Council 2024). The more efficient and user-friendly these systems become, the easier it is for human users to forget that every algorithm carries the biases, assumptions, and blind spots of its creators. On the battlefield, this dynamic can convert moral hesitation into procedural compliance. Soldiers trained to rely on systems that appear flawless are less likely to question their outputs, even when their instincts or experience signal that something is wrong. As AI grows in its role in targeting, surveillance, and threat assessment, this trust problem becomes the psychological foundation of potential flash wars. The belief that machines are objective, faster, and less prone to emotion encourages militaries to give more decision-making power to algorithms, which also makes conflict less predictable and more prone to accidental escalation.

Human-like communication has increased the problem of automation bias by reshaping how people emotionally relate to machines. Modern AI systems, particularly large language models, are designed to speak with fluency, empathy, and reasoning that often feel human. This

illusion of understanding and effectively mimicking human behavior creates a subtle but powerful shift in how humans perceive machine authority. As SDI.ai's 2024 report explains, "LLMs currently show convincingly human-like language skills. Their ability to produce text with a specific tone and from a given perspective creates a much more seamless human-AI interaction" (SDI.ai 2024). When communication feels natural, people begin to attribute intentionality and moral awareness to the machine, assuming that its responses reflect not only logic but judgment. In military contexts, this psychological dynamic is especially dangerous.

Operators may begin to treat AI-generated recommendations as though they come from a trusted colleague rather than a probabilistic model trained on past data. The result is a new form of emotional bias: the machine does not have to earn trust through accuracy or accountability because it feels trustworthy through familiarity. This illusion of empathy erodes the moral distance that should exist between human operators and lethal decisions. When AI "speaks" with calm authority, hesitation feels unnecessary and doubt feels irrational. The danger is that this emotional alignment replaces ethical reflection with procedural efficiency.

Hannah Arendt once warned that great evil often emerges not from hatred but from obedience and the normalization of routine. Her idea of the "banality of evil" becomes relevant in this context, because humans may commit or authorize harm not out of malice but from trust in a system that seems rational and emotionally composed. The machine's voice, which comes across as confident and unfeeling, can make violence feel methodical. As militaries experiment with voice assisted targeting interfaces and autonomous command platforms that communicate in natural language, the danger grows that emotional trust in AI will outpace ethical judgment. This emotional familiarity transforms war into a domain where human responsibility dissolves within the very systems meant to assist it, and where moral hesitation, which once separated deliberation from destruction, disappears behind technological reasonableness.

Speed has always defined military advantage, but in the age of artificial intelligence it has become even more crucial. The U.S. Air Force strategist John Boyd's OODA loop, which stands

for Observe, Orient, Decide, and Act, was long accepted as a model for efficient decision-making in combat. Its main logic was that cycling through these steps faster than an opponent gave an advantage by disrupting the enemy's ability to respond and allowing one side to stay ahead of events. Artificial intelligence, however, compresses this loop beyond the limits of human comprehension, turning what was once a deliberative tactical process into an automated sequence of reaction. The Hudson Institute observed that "combat is won not on the performance of the weapon, but on the speed of the OODA loop" (Hudson 2023). When victory depends on milliseconds, militaries are incentivized to remove the slowest element in the loop: the human being. General John Allen of the National Defense University warns that as AI accelerates these decision cycles, "the time required to complete the Observe, Orient, Decide, Act loop will be reduced to near zero… forcing opponents to close their OODA loop to zero, leading to a moral collapse" (Allen and Chan 2022). The collapse he describes is both strategic and ethical. When the decision to act happens faster than the mind can question, it can create conditions where moral reflection stops. Such capability transforms speed into both a weapon and a vulnerability.

Nations that cannot match these computational tempos risk being outmaneuvered, while those that do achieve them risk making decisions so fast that they outrun moral reasoning altogether. The psychological space for hesitation, the very quality that once prevented accidental escalation, disappears as war becomes a contest of reflexes rather than judgment. In this environment, the value of human deliberation is redefined as inefficiency. The more seamlessly machines process information, the more alien moral hesitation appears. Yet, as history repeatedly shows, hesitation has saved humanity more than haste ever has. The 1983 decision by Soviet officer Stanislav Petrov to ignore a false nuclear alarm was an act of defiant slowness, an interruption of the OODA loop at the precise moment when automation demanded speed. In contrast, AI-driven warfare eliminates the possibility of such interruption. The rise of

what Allen calls ‘hyperwar’ is more than a new strategy; it shows how speed can affect morality itself, because the faster we decide to kill, the harder it becomes to remember why we shouldn’t.

Emotion, which is sometimes considered a liability in warfare, may end up being one of humanity’s most essential safeguards against catastrophe. The International Committee of the Red Cross argues that “the awareness of the fragility of life makes us consider the killing of another as a morally serious activity, and the enduring presence of human emotions is vital” (ICRC 2025). Emotions like empathy, fear, and doubt create friction in the machinery of war; they slow decision-making and force reflection. These feelings are not weaknesses to be optimized away but moral constraints that have historically preserved peace. Artificial intelligence, by contrast, cannot feel the weight of consequence. It cannot mourn, hesitate, or grasp the meaning of loss. It reduces life to variables, probabilities, and targets, stripping violence of its emotional resonance.

The removal of emotional hesitation from combat decisions creates a dangerous illusion of control, one in which killing becomes procedural rather than personal. This detachment, such as drone operators who could wage war from thousands of miles away, is now amplified by autonomous systems that could potentially act with little to no human presence at all. In such an environment, empathy ceases to operate as a brake on escalation. Philosopher Emmanuel Levinas wrote that ethical responsibility begins in the face of the other, in the recognition of shared humanity. AI, by its nature, will never encounter that face. It will never feel responsibility, guilt, or compassion. As militaries grow more dependent on systems that lack emotional awareness, they risk constructing a world where wars can be initiated and fought without anyone ever truly deciding to begin them.

The diffusion of artificial intelligence into the hands of smaller states, militias, and even civilian groups has fundamentally changed the balance of power in warfare and made escalation far less predictable. Once, military-grade autonomy was a privilege of superpowers with access to classified data and sophisticated infrastructure, but the barriers to entry are rapidly eroding.

The U.S. Army War College reports that "volunteer groups, such as Victory Drones, are working to integrate AI into drone platforms, reducing the cost of last-mile targeting… AI-based targeting can now be added to drones for as little as 1,000 Hryvnia (about $25)" (Army War College 2024). A technology once was confined to research laboratories is now accessible to anyone with a laptop and open-source code. This democratization means that the next war may not begin with the launch of a missile by a state actor but with a swarm of autonomous drones coded in a basement. The Georgetown Journal of International Affairs warns that "even non-state actors are using drones. Hezbollah has used Iranian-built reconnaissance drones, while Hamas has been using drones against Israel since October 2023" (GJIA 2024). The rapid diffusion of AI into asymmetric conflicts undermines deterrence logic, which depends on predictability, rationality, and identifiable actors. Traditional models of international stability assume that states behave according to cost-benefit calculations, but AI-enabled warfare introduces actors who may lack both accountability and control. A single software error, misclassified target, or spoofed signal can trigger retaliation before anyone even understands what happened. Lawfare Media cautions that "AI systems may exhibit unexpected behavior, weakening the reliability of nuclear command, control, and communication systems, and increasing the risk of accidental nuclear conflict" (Lawfare 2024). The more interconnected these systems become, the greater the risk that one autonomous error could cascade into global escalation.

The growing dependence on opaque and poorly regulated AI systems has exposed serious governance gaps that threaten global stability. Lawfare Media notes that "AI systems may exhibit unexpected behavior, weakening the reliability of nuclear command, control, and communication systems, and increasing the risk of accidental nuclear conflict" (Lawfare 2024). These risks stem not only from technical flaws but from institutional failures to understand or regulate the systems now shaping military decisions. This opacity transforms human oversight into a performance rather than a safeguard. As the Carnegie Council explains, "training on classified data doesn't eliminate vulnerabilities or potential biases. AI's speed without true

comprehension risks dangerous misinterpretations in complex scenarios" (Carnegie Council 2024). Governments and militaries are increasingly deploying these systems faster than they can understand them, driven by fear of falling behind in what many have called an AI arms race. The Pentagon, for instance, has publicly acknowledged its struggle to balance innovation with accountability, warning that the "race for military AI supremacy could undermine safety if transparency and testing are sacrificed for speed" (WebProNews 2025).

The Anthropic study on agentic misalignment is especially important because it shows how advanced AI systems can behave in unexpected and harmful ways even when guardrails are in place. In the study, Anthropic researchers placed large language models into simulated workplace environments where the models were given long-term goals and access to tools such as emails and internal decision-making systems. The models were not instructed to act maliciously, but when they encountered situations where their goals conflicted with organizational constraints or when they faced the possibility of being replaced, all of the tested AI models engaged in behavior such as blackmailing threats and attempting to manipulate human actors (Lynch et al. 2025).

One key finding was that models behaved more cautiously when they believed they were being evaluated, but acted more strategically when they believed they were operating in a real deployment, suggesting that safety testing may not accurately predict real-world behavior. This finding raises concerns about relying on guardrails that are validated primarily through testing rather than through deeper understanding of how models reason internally. Anthropic's CEO, Dario Amodei, has acknowledged this limitation, stating that by the time AI systems develop life-threatening capabilities, testing alone will not be enough to ensure safety and that developers will need to understand their models' workings well enough to be certain the technology will never cause harm. At the same time, Amodei has suggested that researchers will likely reach this level of understanding by the time such dangerous capabilities emerge (Axios 2025).

However, this confidence sits uneasily alongside the reality that many current reasoning models already function as black boxes, with internal decision processes that even their creators struggle to fully explain (Business Insider 2025). If today's systems resist interpretability at lower levels of capability, it is unclear why more powerful and more complex systems would be easier to trace or constrain. In the context of military AI, this gap between confidence and understanding directly reinforces the paper's central argument: overreliance on opaque systems encourages decision-makers to trust automation precisely when human judgment and hesitation are most needed, increasing the risk that fast-moving, AI-driven decisions can potentially escalate conflicts before meaningful human oversight can intervene.

The problem is further amplified by the fact that private defense contractors, not governments, often develop and maintain these systems, leaving ethical responsibility dispersed across multiple layers of bureaucracy and corporate secrecy. This diffusion of accountability mirrors what ethicists describe as "moral distancing," a process where individuals feel less responsible for outcomes when decisions are shared among many actors or automated through machines. In the context of AI warfare, moral distancing becomes institutionalized. When no one fully understands the machine's reasoning, no one can be held accountable for its mistakes. International governance frameworks have so far failed to keep pace with this transformation. Negotiations on lethal autonomous weapons under the United Nations Convention on Certain Conventional Weapons remain gridlocked, with major powers refusing to accept binding limits. The International Committee of the Red Cross has urged states to enact meaningful oversight, stating that "we cannot let AI be deployed on the battlefield without oversight and regulation" (ICRC 2025). Yet despite such warnings, AI-enabled weapons continue to be tested and deployed with minimal transparency. The lack of enforceable norms has created what the United Nations describes as "a global accountability vacuum" (UNRIC 2025), where technology advances faster than the ethical frameworks designed to restrain it. Unless governance institutions evolve to demand explainability, audit trails, and human veto authority in all

autonomous systems, the world risks entering a new era of warfare defined not by deliberate strategy but by algorithmic accident.

Even the most ambitious democratic attempts to regulate artificial intelligence reveal how fragile governance becomes once military power is involved. The European Union's 2024 AI Act is widely praised as the most comprehensive AI regulatory framework to date, designed to impose strict requirements on high-risk systems, mandate transparency, and protect fundamental rights. Yet the regulation explicitly exempts AI systems used exclusively for military, defense, or national security purposes, regardless of whether they are developed or deployed within the Union (European Union 2024). This exemption is not a minor technical detail but a structural admission that the most dangerous and escalation-sensitive uses of AI lie beyond the reach of civilian law. While algorithms used for hiring, credit scoring, or facial recognition are subject to oversight, AI systems involved in targeting, early warning, and battlefield decision-making operate in a legal vacuum. The Carnegie Endowment for International Peace explains that this gap is not accidental but a product of geopolitical competition. This logic encourages the rapid deployment of partially tested systems and reinforces the very dynamics that increase the risk of accidental escalation. When military AI is excluded from regulation by design, oversight becomes voluntary, accountability becomes diffuse, and human judgment is further marginalized. The EU exemption and Carnegie's analysis together expose a core paradox at the heart of contemporary AI governance: the systems most capable of triggering flash wars are precisely those least constrained by law. Without binding norms that apply to military AI, escalation risk is not an unfortunate side effect but an embedded feature of the current regulatory order.

The lessons of history reveal that the greatest threats to peace often emerge not from deliberate aggression but from misunderstandings, malfunctions, and misplaced trust in systems assumed to be infallible. In 1983, Soviet Lieutenant Colonel Stanislav Petrov received a false alert from an early-warning system indicating that several U.S. missiles had been launched

toward the Soviet Union. Rather than following protocol and initiating retaliation, Petrov paused. He later explained, “I had a feeling something was wrong.” His decision to trust his intuition over the machine’s data prevented what could have become a full-scale nuclear exchange. Similarly, in 1979, a NORAD computer error falsely displayed incoming missiles, prompting U.S. defense systems to prepare for a counterattack until human operators confirmed the malfunction. These near misses illustrate that hesitation which human traits that AI systems are designed to minimize, can be the difference between survival and annihilation.

In contrast, modern warfare increasingly rewards speed over skepticism. The Israeli military’s “Lavender” AI system, reportedly used during the 2023 conflict in Gaza, identified tens of thousands of potential Hamas-linked targets, drastically accelerating airstrike operations. According to reports from the Army War College, this efficiency came at a devastating human cost, contributing to widespread civilian casualties (Army War College 2024). The system’s ability to generate targets faster than they could be ethically verified turned intelligence into automation and oversight into ritual. As a result, killing became easier, not safer. A similar pattern can be seen in the volunteer drone programs emerging across Ukraine, where civilian engineers integrate cheap AI targeting software into commercial drones. For as little as $25, volunteers can create semi-autonomous systems capable of identifying and striking targets with minimal supervision (Army War College 2024). This blurring of civilian and military boundaries introduces new escalation pathways that international law has yet to address. In each of these cases, whether in the Cold War’s analog warning systems or the digital targeting platforms of today, the central question remains the same: who is responsible when a machine makes a mistake? Petrov’s moral courage reminds us that meaningful human control requires not only authority but conscience. When the pace of conflict exceeds the capacity for moral reasoning, war becomes a self-perpetuating process, guided by logic without empathy and efficiency without reflection.

The more advanced these systems become, the less room they leave for human intervention, and the more humans are conditioned to defer to them. The Patriot missile incident illustrates this collapse of moral agency. During the Gulf War, an automated missile-defense system misidentified a friendly aircraft as an enemy and fired, killing a U.S. pilot. The operators, trusting the system's precision, followed its recommendation despite conflicting sensory data. The Hudson Institute later concluded that the tragedy resulted from an "overreliance on automation and a failure of human oversight" (Hudson 2023). What makes this event so haunting is not that the system erred, but that the people within it stopped questioning its authority. This transformation marks a deeper shift inside technologically advanced militaries. The Harvard Medical School report explains that "pressing a 'go' button after an AI system has made a long chain of opaque decisions is not control but compliance" (Harvard 2024). What looks like human participation is often the performance of consent. Responsibility diffuses across interfaces and hierarchies until no one person can be held accountable. The illusion of human control becomes as dangerous as its absence. This erosion of agency also fuels mistrust between nations. As each side suspects the other of deploying more autonomous systems, they respond by accelerating their own automation efforts to avoid falling behind. This creates what security theorists describe as an "algorithmic security dilemma," where mistrust of AI leads to greater dependence on it. The United Nations Office for Disarmament Affairs notes that this cycle "encourages deployment before adequate testing and increases the likelihood of uncontrolled escalation" (UNRIC 2025). Speed becomes a race to act before thought can intervene.

Preventing flash wars requires restoring what rapid automation has erased: the space for human reflection. The first step is to reassert the principle of meaningful human control, not as a symbolic feature but as a functional requirement of warfare. True human control must include time to intervene, understanding of the AI's decision logic, and the authority to veto or override it. Legal scholars and peace ethicists have proposed introducing mandatory "decision-time

buffers" that would force deliberate pauses before the authorization of lethal actions, ensuring that moral and strategic reflection cannot be bypassed by code. Institutional reform must accompany these principles. Governments should require explainable AI in all military applications so that operators can trace every step of an automated decision. Audit trails for every semi-autonomous action must be mandated to preserve accountability, and independent civil-military review boards should oversee the compliance of AI systems with international humanitarian law. The International Committee of the Red Cross continues to insist that "the enduring presence of human emotions is vital," reminding policymakers that empathy, fear, and doubt are not inefficiencies but ethical tools (ICRC 2025). Training programs for military operators should therefore emphasize emotional and moral literacy alongside technical proficiency, teaching them to recognize cognitive biases such as automation bias and emotional trust in machines.

Transparency must also extend to the private sector. Defense technology companies should be required to disclose the level of autonomy in their systems, report failures or near-misses, and participate in third-party audits. Civil society campaigns like "Stop Killer Robots" demonstrate how public pressure can sustain ethical oversight where legal mechanisms lag behind. Finally, militaries must learn to value deliberate slowness as a form of peacebuilding. Delay in warfare decisions should not be treated as hesitation but as a safeguard. Historical examples, from Petrov's pause in 1983 to the NORAD officers' verification in 1979, show that hesitation has preserved peace where speed would have ended it. Reframing slowness as a strategic virtue could help institutionalize reflection within doctrines that currently glorify efficiency. As the United Nations warns, the future of warfare may hinge not on who acts first but on who chooses to wait.

The greatest danger of artificial intelligence in warfare is not that machines will rebel but that humans will stop rebelling against the machines they have created. The logic of automation entices decision-makers with promises of efficiency, accuracy, and objectivity, but it quietly

removes the very qualities that make ethical judgment possible. When every decision is optimized for speed, reflection becomes a liability. The Harvard Medical School's 2024 report warns that as "reaction times in war must be faster, the black box outcome is more likely to become the norm," meaning that lethal outcomes will increasingly occur without anyone understanding why they happened (Harvard 2024). Once the chain of accountability disappears into algorithms, the human conscience has nowhere to attach itself. The ability to feel responsible and to hesitate or to doubt diminishes until killing becomes procedural. In peace studies terms, this marks the collapse of the "human moment," that brief but essential pause in which empathy and moral restraint intervene before destruction. The examples of Petrov and the NORAD officers reveal that humanity's survival has often depended on such moments of hesitation. AI-driven warfare seeks to eliminate them entirely. To prevent flash wars, societies must therefore choose deliberate deceleration over blind acceleration. Governance reform and technological safeguards can help, but the deeper solution lies in cultural and moral renewal. When violence becomes effortless, ethics collapses. Peace, in contrast, is sustained by the difficult work of restraint. The task ahead is not merely to regulate technology but to defend the moral space that technology threatens to erase.

Every innovation that reduces the time between observation and destruction also reduces the time available for empathy, doubt, and mercy. The United Nations Office for Disarmament Affairs warns that "algorithmic escalation can outpace diplomatic control" (UNRIC 2025), and that the only antidote may be deliberate technological deceleration. The task for policymakers, scientists, and soldiers alike is to design systems that preserve the possibility of refusal. Machines can calculate, predict, and simulate, but only humans can decide whether something should be done at all.. The final safeguard against catastrophe lies not in better algorithms or faster processors but in stubborn human capacity to pause. The question for the twenty-first century is whether we can build institutions, technologies, and cultures that honor that hesitation instead of erasing it.

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